

ADAPTIVE IMPEDANCE TOKENS: BRANCH-STRUCTURED EVIDENCE FOR EMBODIED WORLD-ACTION MODELS

Anonymous authors

Paper under double-blind review

ABSTRACT

We study impedance control and robot foundation models. The motivating bet is: Represent impedance choices as adaptive action tokens grounded in contact outcomes. We introduce a mechanism-level representation that keeps physically admissible but unrealized action outcomes as explicit branches. A synthetic contact-style evaluation shows that the proposed branch mechanism improves hidden-mode success and tail-risk relative to observed-only, augmentation-as-data, and scalar-uncertainty baselines.

1 INTRODUCTION

Robotics papers often collapse unobserved physical alternatives into extra data, variance, or a single predicted next state. For impedance control and robot foundation models, this can erase the difference between modes that are visually similar but action-critical. This paper asks whether a world-action model should expose those alternatives to the planner before execution.

2 HOSTILE PRIOR WORK

The closest hostile records include (pri, 2021; 2026; 1997; 2020; 2015; 2018). They make generic world models, contact prediction, and uncertainty insufficient as novelty claims. Our boundary is narrower: preserve branch-valued contact futures as the planning object.

3 MECHANISM

Given state s and action a , the model returns an observed next-state prediction plus a finite atlas $B(s, a)$ of admissible latent branches. Planning minimizes nominal loss plus adverse-branch tail risk. This is not bigger data or a new benchmark; it changes the object represented by the WAM.

4 EVIDENCE

We include a runnable synthetic testbed in `src/run_experiment.py`. It compares four variants under nominal and hidden-mode-shift conditions.

Variant	Split	Success	Tail risk
observed_only_wam	nominal	0.515	0.485
observed_only_wam	hidden_mode_shift	0.180	0.820
augmentation_as_data_wam	nominal	0.603	0.397
augmentation_as_data_wam	hidden_mode_shift	0.260	0.740
scalar_uncertainty_wam	nominal	0.662	0.338
scalar_uncertainty_wam	hidden_mode_shift	0.383	0.617
proposed_branch_mechanism	nominal	0.780	0.220
proposed_branch_mechanism	hidden_mode_shift	0.560	0.440

5 LIMITATIONS

The evidence is synthetic. It supports the mechanism boundary and failure mode, not real-robot state of the art. Hardware validation, richer contact geometry, and learned branch discovery remain open.

6 CONCLUSION

Adaptive Impedance Tokens argues that embodied world-action models should preserve unrealized physical alternatives when those alternatives change downstream action value.

REFERENCES

- Force tracking in impedance control, 1997. <https://doi.org/10.1177/027836499701600107>.
- Stable robust adaptive impedance control of a prosthetic leg, 2015. <https://doi.org/10.1115/dscc2015-9794>.
- Adaptive variable impedance control for dynamic contact force tracking in uncertain environment, 2018. <https://doi.org/10.1016/j.robot.2018.01.009>.
- Smooth adaptive hybrid impedance control for robotic contact force tracking in dynamic environments, 2020. <https://doi.org/10.1108/ir-09-2019-0191>.
- A hybrid control strategy for grinding and polishing robot based on adaptive impedance control, 2021. <https://doi.org/10.1177/16878140211004034>.
- Adaptive admittance-impedance switching control for safe mobile manipulation: A door opening application with predictive anti-slip logic, 2026. <https://doi.org/10.22214/ijraset.2026.79103>.